A Replicable Framework and Open Tool for Assessing Thermal Stress (DHD) in Seagrass Meadows: *Posidonia oceanica* in Balearic Islands Case Study

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1. Importance of the Indicator for the Mar Balear Report

Posidonia oceanica meadows constitute the most emblematic marine ecosystem of the Balearic archipelago, providing critical ecosystem services including oxygen production, carbon sequestration, and coastal protection (Campagne et al., 2015). Marine heatwaves (MHWs), defined as discrete periods of anomalously warm sea surface temperatures exceeding the 90th percentile of local climatology for at least five consecutive days (Hobday et al., 2016), have increased in frequency, duration, and intensity globally over recent decades, with particularly pronounced trends in the Mediterranean Sea (Juza et al., 2022). In the Balearic Sea, the number of marine heatwave days increased by 114 days over the same period (Fernández-Álvarez et al., 2025). However, translating these physical metrics into actual biological risk for the ecosystem remains challenging. To assess the environmental extent of this phenomenon, we developed a spatially explicit indicator that converts thermal anomalies into biological risk for the viability of this iconic species. This indicator identifies meadows where physiological capacity has been exceeded, initiating net regression. It also highlights areas that maintain structural resilience but suffer metabolic imbalance, a condition that could lead to future fragmentation and eventual decline (Giménez-Romero et al., 2026). Both scenarios call for urgent measures to reduce local pressures and enhance meadow resilience.

2. Data Acquisition and Modelling Methodology

The indicator integrates satellite remote sensing, benthic cartography, and in situ empirical calibration through an advanced statistical modelling approach:

A. Quantification of Thermal Threat (DHD). We used daily Sea Surface Temperature (SST) raster data from the Copernicus Marine Service (CMEMS). To capture cumulative stress, we calculated the Degree-Heating-Days (DHD) index by summing daily thermal anomalies above the acute stress threshold of 28.0 °C (Marbà & Duarte, 2010) during the summer season from 2017 to 2024.

B. Spatial Integration. The analysis was strictly limited to the actual distribution of the meadows using the official IDEIB cartography (Layer 2: *Posidonia oceanica*). We incorporated bathymetry as a modulating covariate because thermal impact attenuates with depth.

C. Empirical Calibration via GAMM. To ensure scientific robustness, we calibrated the indicator with global shoot density data from the Xarxa de Monitoratge de la Posidònia (Govern de les Illes Balears, 2017, 2019, 2022, 2023, 2024). We fitted a Generalised Additive Mixed Model (GAMM) with a scaled t-distribution, which models interannual density change as a function of accumulated DHD and depth, while random effects account for variability among meadows and stations. We selected the scaled t-distribution for its capacity to accommodate the high variability observed in the contemporary Mediterranean, including mass mortalities and high compensatory growth rates. The model was validated through spatial cross-validation and residual diagnostics, explaining 20% of the deviance, with both accumulated heat (DHD) and depth as significant predictors. From the model curve, we derived two operational thresholds. The critical threshold (5.6 DHD) corresponds to the zero-crossing of the prediction curve, the point at which the model predicts net meadow regression. The alert threshold (4.9 DHD) was defined as the 75th percentile of DHD values recorded at stations that had not yet reached the critical threshold, representing the upper limit of the sub-lethal stress range documented in the field.

3. Results and Observed Trends

The model identified an empirical regional inflection point (5.6 DHD) beyond which shoot mortality exceeds regeneration, causing net meadow regression. Based on this calibration, the marine space is classified into three thermal vulnerability states: Green Zone (Stabilisation, < 4.9 DHD), Yellow Zone (Sub-lethal stress, 4.9-5.6 DHD), and Red Zone (Net regression, > 5.6 DHD).

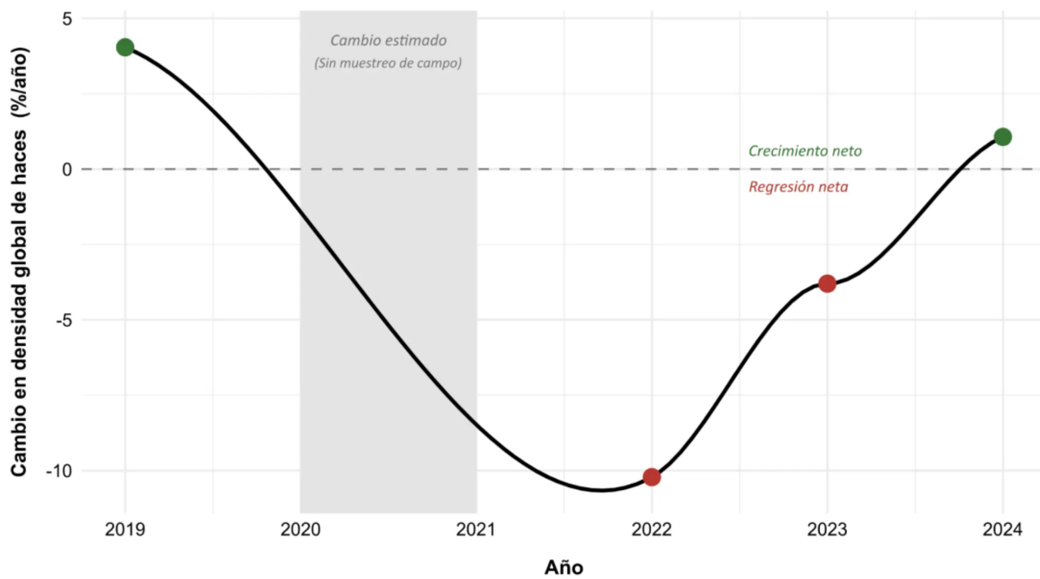

Figure 1. Temporal evolution of thermal impact on global shoot density. The model reveals that, even before the extreme 2022 heatwave, the meadows were already experiencing a trajectory of net regression due to accumulated stress, demonstrating that chronic warming acts as a silent but constant stressor.

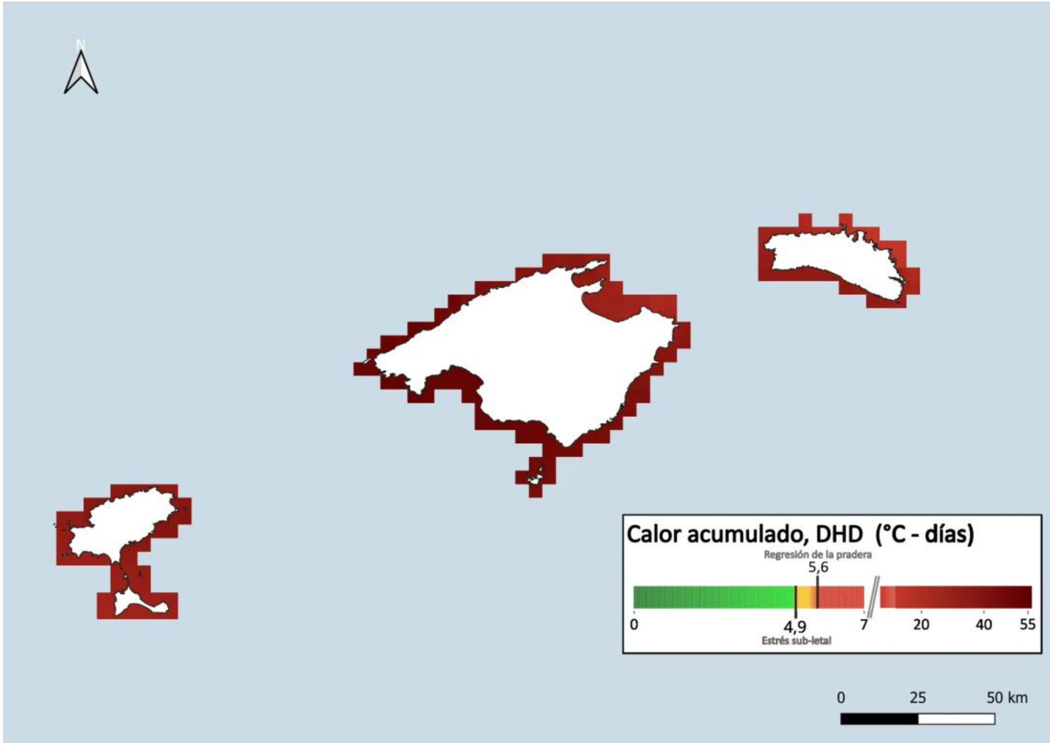

Figure 2. Thermal Stress Map for *Posidonia oceanica* in 2022. During the 2022 heatwave, nearly all Balearic meadows exceeded the critical threshold of 5.6 DHD (dark red colour), entering a state of widespread biological deficit.

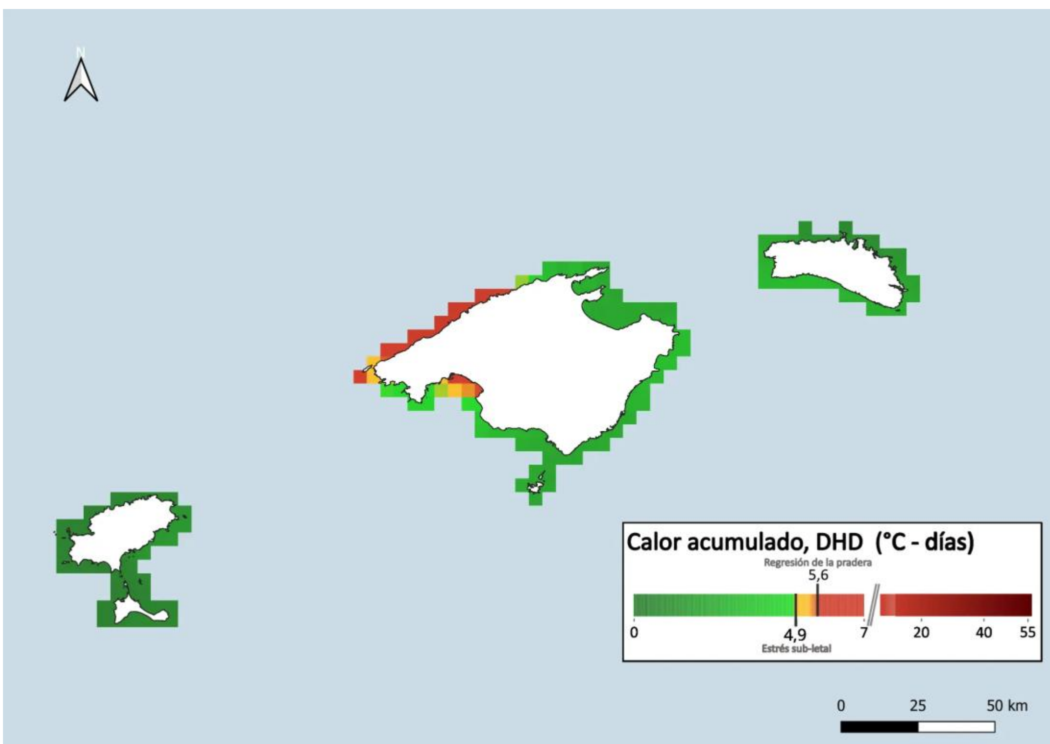

Figure 3. Thermal Stress Map for *Posidonia oceanica* in 2024. In a year without an officially designated extreme heatwave, the model reveals marked spatial variability. Although most of the archipelago remains in the safe zone (green), localised areas (e.g., northwest Mallorca) still exceed the alert and regression thresholds, indicating that thermal stress is spatially heterogeneous and may be recurrent in certain zones.

4. Discussion and Relevance for Marine Management

This indicator transforms abstract climatic data into a spatial roadmap for management. Because marine heatwaves cannot be controlled at a local scale, the functional survival of Posidonia depends on maximising its resilience by reducing other human pressures.

The model does not measure whether a year was warm in absolute terms, but whether the accumulated heat exceeded the physiological threshold of the plant. Between 2019 and 2021, chronic stress was already slowly eroding the meadows, though without reaching collapse. In 2022, the heatwave pushed them to the limit. The 2024 data provide an encouraging signal: Balearic meadows are demonstrating resilience. Following the extreme impact of 2022, the indicator records net growth at a regional level for the first time in 2024, confirming that the ecosystem can recover when thermal conditions improve. The indicator captures this dynamic precisely: not only the acute peaks, but also the cumulative wear and the subsequent rebound, albeit from a much lower baseline density than in 2019.

However, the 2024 map reveals that this recovery is not uniform. While most of the archipelago returns to the green zone, localised areas, particularly in northwest and southwest Mallorca, continue to exceed alert or regression thresholds even in a year without an extreme heatwave. These areas require urgent management to reduce local pressures and allow recovery. The map facilitates this targeted action:

1. Identification of Climate Refugia. Areas that consistently remain in the Green Zone represent the climate refugia of the archipelago. These must be prioritised for continuous monitoring by the *Xarxa de Monitoratge* and, given their good conservation status, they constitute the most suitable genetic reservoirs for potential restoration actions in degraded areas. Management in these areas does not require additional restrictions, but should ensure that existing pressures do not increase.

2. Management of Synergistic Stress. Meadows in the Yellow and Red Zones are at or beyond the limit of their physiological tolerance. In Red Zone areas, particularly where the impact is recurrent, authorities should implement Zero Tolerance policies for manageable local stressors, including strict anchoring controls and regulation of discharges of boats and submarine outfalls. Reducing these pressures is the most effective and directly actionable management pathway to prevent synergistic stress from precipitating ecosystem collapse. In Yellow Zone areas, which have not yet collapsed, proactive monitoring and preventative reduction of local impacts are essential to halt progression towards net regression.

References

Campagne, C. S., Salles, J.-M., Boissery, P., & Deter, J. (2015). The seagrass *Posidonia oceanica*: Ecosystem services identification and economic evaluation. *Marine Pollution Bulletin*, *97*(1–2), 391–400. https://doi.org/10.1016/j.marpolbul.2015.05.061

Fernández-Álvarez, B., Juza, M., & Tintoré, J. (2025). Tracking marine heatwaves in the Balearic Sea. *EGUsphere* [Preprint]. https://doi.org/10.5194/egusphere-2024-4065

Giménez-Romero, À., Matías, M. A., Hendriks, I. E., & Duarte, C. M. (2026). Thermal stress is associated with fragmentation of Mediterranean *Posidonia oceanica* meadows. *Global Change Biology Communications*, *1*(3), Article e70031. https://doi.org/10.1002/gcb4.70031

Govern de les Illes Balears. (2017). *Memòria de la Xarxa de Monitoratge de les Praderies de Posidònia 2017*. Direcció General de Medi Natural i Gestió Forestal, Conselleria de Medi Ambient i Territori. https://www.caib.es/sites/posidonia/ca/definicia/

Govern de les Illes Balears. (2019). *Memòria de la Xarxa de Monitoratge de les Praderies de Posidònia 2019*. Direcció General de Medi Natural i Gestió Forestal, Conselleria de Medi Ambient i Territori. https://www.caib.es/sites/posidonia/ca/definicia/

Govern de les Illes Balears. (2022). *Memòria de la Xarxa de Monitoratge de les Praderies de Posidònia 2022*. Direcció General de Medi Natural i Gestió Forestal, Conselleria de Medi Ambient i Territori. https://www.caib.es/sites/posidonia/ca/definicia/

Govern de les Illes Balears. (2023). *Memòria de la Xarxa de Monitoratge de les Praderies de Posidònia 2023*. Direcció General de Medi Natural i Gestió Forestal, Conselleria de Medi Ambient i Territori. https://www.caib.es/sites/posidonia/ca/definicia/

Govern de les Illes Balears. (2024). *Memòria de la Xarxa de Monitoratge de les Praderies de Posidònia 2024*. Direcció General de Medi Natural i Gestió Forestal, Conselleria de Medi Ambient i Territori. https://www.caib.es/sites/posidonia/ca/definicia/

Hobday, A. J., Alexander, L. V., Perkins, S. E., Smale, D. A., Straub, S. C., Oliver, E. C. J., Benthuysen, J. A., Burrows, M. T., Donat, M. G., Feng, M., Holbrook, N. J., Moore, P. J., Scannell, H. A., Sen Gupta, A., & Wernberg, T. (2016). A hierarchical approach to defining marine heatwaves. *Progress in Oceanography*, *141*, 227–238. https://doi.org/10.1016/j.pocean.2015.12.014

Juza, M., Fernández-Mora, À., & Tintoré, J. (2022). Sub-regional marine heat waves in the Mediterranean Sea from observations: Long-term surface changes, sub-surface and coastal responses. *Frontiers in Marine Science*, *9*, Article 785771. https://doi.org/10.3389/fmars.2022.785771

Marbà, N., & Duarte, C. M. (2010). Mediterranean warming triggers seagrass (*Posidonia oceanica*) shoot mortality. *Global Change Biology*, *16*(8), 2366–2375. https://doi.org/10.1111/j.1365-2486.2009.02130.x